2023-Winter Paper 1 – Basic Sciences in Urology

Contents

1. Q1. Principles of tissue engineering and its use in urology (LAQ)
2. Q2. Enlist various grafts used in urology and role of BMG in urology (LAQ)
3. Q3a. Management of perforated mid urethral sling (SAQ)
4. Q3b. Robotic radical prostatectomy (SAQ)
5. Q3c. Kidney retrieval techniques for organ donation (SAQ)
6. Q3d. Diagnosis and management of hemorrhagic cystitis (SAQ)
7. Q3e. Surgical management of primary scrotal cancer (SAQ)
8. Q3f. Female circumcision (SAQ)

Q1. Principles of tissue engineering and its use in urology

Type: LAQ (25 marks) (also answered in 2022-Summer Paper 4, Q2; this answer stresses the principles)

Definition and rationale

- **Tissue engineering** applies engineering and life sciences to build biological substitutes that restore, maintain or improve tissue function (Langer and Vacanti, 1993).
- **Why in urology:** bowel in the urinary tract causes mucus, stones, metabolic acidosis, infection and cancer; grafts for long urethral defects are scarce. The aim is to replace "wrong tissue" with "right tissue".

Principles: the triad plus bioreactor and blood supply

```
Biopsy -> cells expanded in culture -> seeded on scaffold (+ signals)
       -> conditioned in bioreactor -> implanted -> host vascularises and remodels
```

1. Cells

Source	Comment
Autologous differentiated cells (urothelial and smooth muscle cells from a bladder biopsy)	No rejection; expand well; not usable if the organ is malignant
Adult stem cells: bone marrow or adipose mesenchymal, muscle-derived, urine-derived	Easy harvest; act largely by paracrine (growth factor) effects
Embryonic stem cells	Pluripotent; ethical issues, teratoma risk
iPSCs (induced pluripotent stem cells)	Adult cells reprogrammed; patient-specific; tumour risk

2. Scaffolds (biomaterials)

- **Requirements:** biocompatible; degrade at the rate new tissue forms; porous for cell entry and vessel ingrowth; mechanically suited (a compliant bladder, a non-collapsing tube).

Type	Examples	Feature
Natural polymers	Collagen, alginate, silk	Good cell recognition; weak
Acellular tissue matrices	Bladder acellular matrix, SIS (small intestinal submucosa)	Native architecture and stored growth factors

Type	Examples	Feature
Synthetic polymers	PGA (polyglycolic acid), PLA (polylactic acid), PLGA (their copolymer)	Controlled shape and degradation; no bioactivity
Newer	Electrospun nanofibres, hydrogels, bio-ink	Mimic extracellular matrix

- **Seeding:** urothelial cells on the luminal side and smooth muscle cells on the outer side, copying native layers.
- **Unseeded** matrices rely on host cells migrating in, so they work only for short onlay defects (about 0.5 cm for tubes in animal work). Longer or tubular gaps need **seeded** scaffolds, or they fibrose and contract.

3. Signals

- Growth factors (for example VEGF, vascular endothelial growth factor, for vessels) released from the scaffold, extracellular-matrix cues and mechanical stimuli direct cell growth and differentiation.

4. Bioreactors

- Culture chambers that perfuse nutrients and oxygen through thick constructs and apply **physiological conditioning**, such as cyclic stretch and filling for bladder smooth muscle, so the tissue is functional before implantation.
- Types: static culture, spinner flask, rotating wall, perfusion and mechanical-stretch bioreactors. They also allow monitoring and sterile, standardised production.

5. Vascularisation and innervation

- Oxygen diffuses only about 100–200 µm, so thick constructs die centrally without vessels. This is the main barrier.
- Strategies: **omental wrap** (used by Atala), angiogenic factors, co-seeding endothelial cells, built-in channels by bioprinting.
- Innervation and coordinated contraction cannot yet be recreated.

Uses in urology

Organ	Status
Bladder	**Atala et al. (2006):** seeded collagen–PGA constructs with omental wrap in 7 myelomeningocele patients; compliance improved. Industry phase II trial: **no significant gain in compliance or capacity**. Experimental (Chapter 48).
Urinary conduit	Neo-Urinary Conduit trial (NCT01087687): survived about 250 days on average; **half developed stenosis**; stiff material, no blood supply (Chapter 135)
Urethra	Acellular matrix onlay for short strictures and hypospadias; **Raya-Rivera et al. (2011):** seeded tubular PGA–PLGA urethras in **5 boys**, functional up to 6 years; tissue-engineered oral mucosa
Sphincter	Autologous muscle-derived cells for SUI (stress urinary incontinence); biodegradable scaffold slings
Vagina	Engineered vaginas in MRKH (Mayer–Rokitansky–Küster–Hauser) syndrome (Raya-Rivera, 2014)
Penis	Corporal tissue and tunica grafts (stem cell-seeded SIS in Peyronie disease): animal or experimental; stem cells for ED (erectile dysfunction) investigational
Kidney	Decellularised–recellularised scaffolds (animal), kidney organoids, renal assist devices
Testis	Spermatogonial stem cell banking in prepubertal cancer patients

Recent advances

- **3D bioprinting:** cells in hydrogel "bio-ink" laid layer by layer from CT (computed tomography) or MRI (magnetic resonance imaging) data; vascular channels built in. Printed

(non-living) models already aid planning.
- **Organoids:** kidney organoids from iPSCs; bladder and prostate tumour organoids for drug testing.
- Robotic vascularised composite bladder allograft (Chapter 135): a transplant route that avoids bowel but needs immunosuppression.

Challenges

- Vascularity, innervation, fibrosis and contracture; scale-up, cost, culture time, regulation; stem cell tumour risk; promising pilots followed by disappointing controlled trials.
- **Also know:** gene-modified cells; growth factor-releasing "smart" scaffolds; cell sheet engineering.

Diagram to draw: biopsy → culture → seeding on scaffold → bioreactor → implant with omental wrap.

> **Key points to remember:**
> - Triad: cells, scaffold, signals; plus bioreactor and blood supply.
> - Diffusion limit about 100–200 µm: vascularity is the main barrier.
> - Atala 2006 promising; phase II trial showed no gain; conduit half stenosed.
> - Raya-Rivera 2011: seeded urethras in 5 boys.

Q2. Enlist various grafts used in urology and role of BMG in urology

Type: LAQ (25 marks) (BMG uses also in 2015-Summer Paper 3, Q3b; vascular grafts in 2021-Summer Paper 3, Q2)

Graft versus flap

	Graft	Flap
Blood supply	Separated; gets a new supply from the bed	Moved with its own pedicle
Depends on	A well-vascularised host bed	Its pedicle
Example	BMG (buccal mucosal graft)	Penile fasciocutaneous island flap

Graft take (about 96 hours)

Phase	Duration	Mechanism
Imbibition	about 48 h	Graft "drinks" nutrients from the bed; graft is cooler than core
Inosculation	about 48 h	Bed and graft vessels connect; true microcirculation returns

- Take fails with poor beds (scar, radiation), haematoma, seroma, infection or shear. Hence **quilting or pie-crusting** and **immobilisation for 5–10 days** (bolster or vacuum dressing).

Grafts used in urology

Group	Graft	Main uses
Skin	**STSG** (split-thickness skin graft), meshed	Staged urethroplasty only; penile and scrotal cover (Fournier gangrene, buried penis)
	FTSG (full-thickness skin graft)	Fossa navicularis, penile urethra (avoid in lichen sclerosus)
	Posterior auricular (Wolfe) FTSG	Staged hypospadias repair
Mucosa	**Buccal, lingual, labial (oral mucosa)**	Urethra, hypospadias, ureter
	Bladder epithelium	Urethra, hypospadias
	Rectal (colonic) mucosa	Panurethral stricture when oral mucosa is unavailable
Other autologous	Dermis	Tunica albuginea in Peyronie disease
	Tunica vaginalis	Peyronie disease, urethral cover; easy harvest
	Saphenous vein	Peyronie defects; vascular repair
	Rectus fascia, fascia lata	Pubovaginal sling
Allograft / xenograft	Pericardium, porcine SIS, porcine dermis, cadaveric fascia	Peyronie grafting ("off the shelf"); slings
Synthetic	Polypropylene (macroporous type I mesh)	Mid-urethral sling, prolapse
	Dacron, PTFE (polytetrafluoroethylene)	Vascular grafts; not advised for Peyronie (infection, fibrosis)
Newer	Collagen fleece (TachoSil); acellular matrix, tissue-engineered oral mucosa	Peyronie; experimental urethra

- Vein is not a true graft: it survives by direct endothelial perfusion and its vasa vasorum.
- **Peyronie grafts:** pericardium (thin, strong, does not contract) and SIS (cut about 25% oversize; contracts) are commonest; AUA gives no preference.
- **STSG versus FTSG:** STSG carries little dermis, takes easily, can be meshed (up to sixfold) but contracts. FTSG carries full dermis, contracts less, looks better but is more "fastidious" in take.

Role of BMG

Why it works

- Thick, non-keratinised, elastin-rich epithelium used to a **wet environment**; resists infection.
- Thin, **highly vascular lamina propria** with a panlaminar plexus, so imbibition and inosculation are rapid.
- Hairless; little contraction; handles and holds sutures well; low donor morbidity.

Harvest

1. Nasotracheal intubation; mouth gag; mark the graft (often about 4–5 × 2 cm per cheek).
2. Stay 1 cm from the commissure and avoid **Stensen's (parotid) duct** opposite the upper second molar.
3. Infiltrate, incise, raise the mucosa off buccinator; haemostasis.
4. Donor site: leave open or close; both acceptable.
5. Back table: **defat** the graft, cut about 10% oversize, fenestrate with a needle.

- **Lingual** (under the tongue) and **labial** grafts are alternatives. Buccal and lingual give equal urethroplasty success; lip mucosa is thinner, suits the glans.

Uses in urethroplasty

Technique	Principle	Best for
Dorsal onlay (Barbagli, 1995)	Dorsal urethrotomy; graft spread-fixed on corpora	Bulbar; reliable bed
Ventral onlay	Ventral urethrotomy; graft covered by spongioplasty	Bulbar with healthy spongiosum
Dorsal inlay (Asopa)	Ventral approach; dorsal plate incised; graft inlaid	Penile and bulbar
Augmented anastomosis	Excise narrowest part, floor anastomosis, graft roof	Long, dense bulbar stricture
Kulkarni (2012)	One-sided dissection; penile invagination; dorsal graft	Panurethral, one stage
Staged (Johanson, Bracka)	Stage 1 graft; tubularise after 6 months or more	Lichen sclerosus, failed hypospadias
BMG perineal urethrostomy	Graft completes dorsal wall to skin	Panurethral or failed repairs

- Dorsal and ventral results are similar; onlay beats tubes. **Skin is avoided in lichen sclerosus** because the disease recurs in skin.

Other uses

- **Ureter:** BMG onlay ureteroplasty (first human report 1999), now robotic with an **omental wrap**; for long proximal or mid strictures in dense scar. CORRUS (Collaborative of Reconstructive Robotic Ureteral Surgery) failure about 15–25% by site.
- Redo and staged hypospadias; urethral closure in rectourethral fistula; bladder neck contracture; female urethral stricture; Peyronie grafting (small defects only).

Complications

- **Donor:** pain, swelling (first month), numbness (up to 6 months), mouth tightness, salivary duct injury.
- **Recipient:** graft failure, restricture, sacculation, fistula, ED.

Outcomes

- Bulbar BMG urethroplasty about 85–90% success; most failures within 12 months.

Diagram to draw: bulbar cross-section showing dorsal onlay, ventral onlay and dorsal inlay positions.

> **Key points to remember:**
>
> - Take: imbibition 48 h, then inosculation 48 h; graft lives on its bed.
> - BMG: non-keratinised, vascular lamina propria, wet-adapted.
> - Avoid Stensen's duct; buccal equals lingual in success.
> - Staged BMG, not skin, in lichen sclerosus.

Q3a. Management of perforated mid urethral sling

Type: SAQ (10 marks)

Terminology (IUGA/ICS, International Urogynecological Association / International Continence Society)

- **Perforation:** mesh in a hollow organ (bladder, urethra). **Exposure:** mesh seen through unhealed vaginal epithelium. **Extrusion:** passage out of a tissue. "Erosion" is no longer used.
- Two settings: **intraoperative trocar injury** and **late mesh perforation**.

A. Intraoperative trocar injury

- Bladder injury: 4.5% retropubic versus **0.6% transobturator**; registries 2.7–3.8%.
- **Detection:** cystoscopy with a **70° lens and full bladder distension** after every trocar pass.

Injury	Management
Bladder	Withdraw trocar, **repass more laterally**, confirm on cystoscopy, complete the sling; catheter overnight to 72 h. Benign; no later mesh perforation.
Urethra	**Do not place mesh** (AUA/SUFU, American Urological Association / Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction). Repair, catheter drainage; **staged** anti-incontinence surgery after healing.

B. Late perforation of bladder or urethra

- **Presentation:** recurrent UTI (urinary tract infection), haematuria, urgency, pain, stones on mesh, voiding dysfunction, fistula. Often months later.
- **Evaluation:** examination, **cystourethroscopy**, translabial ultrasound (best imaging for mesh); exclude fistula.

Management

- **Bladder:** endoscopic excision (scissors, resection or holmium laser) flush with the wall; or transvesical laparoscopic or robotic excision; open cystotomy if large.
- **Urethra:** transvaginal excision of mesh in the field and layered urethral repair; **Martius flap** if complex or fistula.
- **Continence after excision:** SUI recurs in many (44–100% after synthetic sling revision). Options: concomitant **autologous pubovaginal sling** (Shah 2013: good results) or staged. A damaged urethra predicts poor continence.
- Avoid urethral dilation after mesh slings; it can cause perforation.

Prevention

- Empty bladder; trocar hugging the pubis; bottom-up retropubic route; transobturator route when suitable.

Also know: Amid type I macroporous polypropylene integrates best; chronic pain or infection may need excision from both sides of the bone.

Diagram to draw: retropubic and transobturator trocar paths with bladder at risk.

> **Key points to remember:**
> - Bladder trocar injury: repass laterally, continue.
> - Urethral injury: abandon mesh; stage the repair.
> - Late perforation: excise mesh, repair, consider autologous sling.

Q3b. Robotic radical prostatectomy

Type: SAQ (10 marks)

Introduction

- RARP (robot-assisted radical prostatectomy): Walsh's anatomical nerve-sparing principles done with **3D, 10× magnified vision and wristed instruments**. The robot removed the laparoscopic suturing barrier; da Vinci from 2000 (S, Si, Xi, SP).

Indications and contraindications

- Localised cancer (cT1–T2, selected cT3); life expectancy over 10 years.
- **Absolute:** uncorrectable bleeding diathesis; unfit for general anaesthesia.
- **Relative:** morbid obesity, prostate >80 g, median lobe, prior TURP (transurethral resection of the prostate), salvage surgery.

Set-up

- Low lithotomy, arms tucked, **steep Trendelenburg (22–25°)**, padding; pressure 12–15 mmHg; camera port 15–17 cm above pubis, three robotic and two assistant ports.

Steps (transperitoneal anterior approach)

1. Drop bladder; enter space of Retzius.
2. Incise endopelvic fascia; ligate DVC (dorsal venous complex).
3. Divide bladder neck (fat line, balloon traction); check for median lobe.
4. Seminal vesicles and vasa; incise Denonvilliers fascia.
5. Clip pedicles athermally; **interfascial nerve sparing** (extrafascial if high risk).
6. Apical dissection preserving urethral length.
7. Extended PLND (pelvic lymph node dissection) if indicated.
8. Posterior (Rocco) reconstruction; **Van Velthoven running anastomosis**.

Postoperative care

- ERAS (enhanced recovery after surgery); discharge within 24 h; catheter about 1 week (not before day 5: retention); pelvic floor training; penile rehabilitation with PDE5 (phosphodiesterase type 5) inhibitors.

Outcomes

- **Yaxley 2016** randomised trial: equal function and early margins versus open; less blood loss (300 versus 1400 mL) and shorter stay.
- Margins about 15% (9% if pT2); 1-year continence mid-90s% in expert hands.

Complications

- Transfusion 1–2%; bladder neck stricture <2% (open up to 16%); rectal injury; ureteric injury; lymphocele; positioning neuropathy, corneal abrasion; robot failure <0.5%.

Recent advances

- **Retzius-sparing** (earlier continence, more positive margins); Hood technique; single-port robot; same-day discharge.

Diagram to draw: port layout and fascial planes (intra-, inter-, extrafascial).

> **Key points to remember:**
>
> - Surgeon, not platform, decides outcome (Yaxley 2016).
> - Interfascial plane is the standard nerve-sparing plane.
> - Bladder neck stricture <2%; transfusion 1–2%.

Q3c. Kidney retrieval techniques for organ donation

Type: SAQ (10 marks) (also answered in 2021-Summer Paper 2, Q2, with donor selection)

Types of donor

- **Deceased:** DBD (donation after brain death; circulation maintained) or DCD (donation after circulatory death; warm ischaemia).
- **Living:** related or unrelated (THOA, Transplantation of Human Organs Act, in India).

A. Deceased multi-organ retrieval (DBD)

1. Midline laparotomy with sternotomy.
2. Mobilise bowel off retroperitoneum (Cattell–Braasch); expose aorta and IVC (inferior vena cava); **look for and preserve accessory renal arteries**.
3. Heparin; cannulate distal aorta; clamp supracoeliac aorta; **vent IVC**; flush cold UW (University of Wisconsin) or HTK (histidine–tryptophan–ketoglutarate) solution; surface ice; open Gerota's fascia laterally.
4. Heart, lungs, liver and pancreas removed first.
5. Kidneys removed **en bloc** with aorta, IVC, perinephric fat and ureters, keeping periureteric tissue.

- **DCD:** super-rapid laparotomy and direct aortic cannulation to cut warm ischaemia; usually machine perfusion.
- **Back table:** separate kidneys; aortic **Carrel patch**; IVC lengthens the short right renal vein; ligate lymphatics; biopsy marginal kidneys; package at 4 °C.
- Paediatric donors <15 kg: en bloc kidneys.

B. Living donor nephrectomy

- **Principles:** donor safety first; maximal vessel length; keep periureteric tissue and the lower pole–ureter triangle; minimal handling; short warm ischaemia.
- **Left kidney preferred** (longer vein).

Technique	Features
Open (flank or mini-incision)	Complex anatomy, more often right side
Laparoscopic (Ratner, 1995)	Preferred; less pain, faster recovery
Hand-assisted	About two-thirds of US cases; tactile control
Retroperitoneoscopic, LESS (laparoendoscopic single-site), robotic	Selected centres

- Sequence: hilar dissection → ureter divided → artery then vein stapled or clipped → rapid extraction → cold flush.
- **Risks:** complications about 16%; mortality 3 per 10,000; conversion 0.26–3.34%.

Preservation

- Cooling slows metabolism; hypothermic machine perfusion reduces DGF (delayed graft function), not long-term loss.

Diagram to draw: aortic cannula, supracoeliac clamp, IVC vent, en bloc kidneys.

Key points to remember:

- Cannulate, clamp, vent, cool; kidneys removed last, en bloc.
- Left kidney preferred for living donation.
- Machine perfusion reduces DGF.

Q3d. Diagnosis and management of hemorrhagic cystitis

Type: SAQ (10 marks) (also answered in 2022-Summer Paper 3, Q3b)

Definition

- Diffuse inflammation and bleeding of the bladder mucosa, after other sources are excluded.

Causes

Cause	Mechanism
Cyclophosphamide, ifosfamide	Liver forms **acrolein**, which sloughs urothelium; onset within 48 h
Pelvic radiation (about 5%; 6 months to 10 years)	**Obliterative endarteritis**, ischaemia, fibrosis
Viral	BK polyomavirus; adenovirus 11, 35; after transplant
Others	Bacteria; busulfan, thiotepa; penicillins; amyloidosis

Diagnosis

- History: drugs, radiotherapy, transplant, anticoagulants. Severity: microscopic, gross, clots, clot retention.
- Blood count, coagulation, renal function, urine culture, viral PCR (polymerase chain reaction) if transplanted.
- **Exclude cancer:** cytology, cystoscopy with biopsy, upper tract imaging (CT urography).

Prevention

- **Mesna** inactivates acrolein; hyperhydration, forced diuresis.

Management (escalate by severity)

```
Resuscitate, transfuse, correct coagulopathy, treat infection
→ 3-way catheter + CBI (continuous bladder irrigation)
→ clot evacuation, fulguration, biopsy under anaesthesia
→ intravesical alum 1%
→ bleeding after 48 h: hyperbaric oxygen (+ nephrostomies)
→ formalin 1%, then 4% (cystogram first)
→ cystectomy + conduit diversion
(Unstable at any stage: internal iliac embolisation)
```

Agent	Key point
Alum 1%	First line; no anaesthesia; avoid in renal failure (aluminium toxicity)
Aminocaproic acid	Antifibrinolytic CBI; clot-free bladder only
Prostaglandins, silver nitrate, hyaluronate	Second-line options
Formalin 1–4%	80–90% success; under anaesthesia, instilled by gravity for 10–15 min; **cystogram to exclude reflux** (occlude ureters if present); risks: contracted bladder, ureteric stricture

- **HBOT** (hyperbaric oxygen therapy): 100% oxygen at 2–3 atmospheres, 30–40 sessions; raises tissue oxygen and drives neovascularisation; 80–90% response, but many relapse; **RICH-ART** trial improved urinary scores in radiation cystitis. Nephrostomies divert urokinase-rich urine if bleeding continues.

- **Embolisation:** selective anterior division of internal iliac (posterior division causes gluteal pain); initial control about 90%.
- **Cystectomy** last; remove bladder (pyocystis).
- **Viral:** supportive (hydration, irrigation); cidofovir anecdotal.
- **Also know:** tranexamic acid and laser ablation are unproven.

Diagram to draw: escalation ladder.

> **Key points to remember:**
> - Acrolein (mesna); radiation endarteritis; BK virus.
> - Exclude bladder cancer.
> - Alum first; formalin after cystogram; HBOT for radiation.

Q3e. Surgical management of primary scrotal cancer

Type: SAQ (10 marks) (also answered in 2021-Summer Paper 4, Q3c)

Introduction

- Rare; **SCC (squamous cell carcinoma)** commonest. **Percivall Pott (1775)**: chimney sweeps, the first occupational cancer.
- **Risks:** soot, tar, mineral oils, arsenic, PUVA (psoralen + ultraviolet A), HPV (human papillomavirus).
- **Others:** EMPD (extramammary Paget disease), BCC (basal cell carcinoma), melanoma, sarcoma.
- Drains to **superficial inguinal nodes, crossing the midline**.

Evaluation

- Painless nodule or ulcer on the anterolateral scrotum, often presenting late; **biopsy** confirms.
- Examine both groins (lymphatics cross the midline); FNAC (fine-needle aspiration cytology) of nodes; CT (computed tomography) of abdomen and pelvis for staging.

Staging (Lowe)

Stage	Extent
A1	Confined to scrotum
A2	Invades testis, cord, penis, pubis or perineum
B	Inguinal nodes
C	Pelvic nodes
D	Distant metastases

Primary tumour

1. **Wide local excision, 2 cm margin**, full thickness with dartos; frozen sections.
2. Testes spared unless invaded; hemi- or total scrotectomy for large lesions.
3. A2: en bloc removal of involved testis, penis, perineum or pubis.
4. Positive margin: re-excise.

Reconstruction (Chapter 87)

Defect	Management
<50%, no tension	Primary closure
<50%, tension	Scrotal advancement flap or secondary intention
>50%	Testes in thigh pockets; later STSG (split-thickness skin graft) or flap

Nodes (as in penile cancer)

- **Palpable, cytology positive:** ILND (inguinal lymph node dissection) ± pelvic dissection. Nodal status is the main prognostic factor.
- **Impalpable:** surveillance; sentinel node biopsy in high-grade tumours.
- **Bulky or fixed:** neoadjuvant cisplatin-based chemotherapy, then surgery; RT (radiotherapy) for palliation.
- Routine prophylactic dissection is not advised.

Other types

- **EMPD:** wide excision guided by mapping biopsies; screen for other cancers.
- **Also know:** BCC, excision or Mohs surgery; melanoma, excision by Breslow depth ± sentinel node; sarcoma, wide excision.

Diagram to draw: elliptical excision with 2 cm margin and bilateral inguinal drainage.

> **Key points to remember:**
>
> - Pott 1775: chimney sweeps.
> - 2 cm margin; testes spared.
> - Nodes decide prognosis.

Q3f. Female circumcision

Type: SAQ (10 marks) (also answered in 2019-Summer Paper 2, Q3f)

Taken as female genital mutilation/cutting (Chapter 58 calls it "female circumcision (infibulation)").

Definition and epidemiology

- **FGM (female genital mutilation):** partial or total removal of, or other injury to, the external female genitalia for non-medical reasons (WHO, World Health Organization).
- About **200 million** females affected; about 3 million African girls a year. Africa, Middle East, Indonesia and Malaysia; some Indian communities.
- Usually at 4–10 years, without anaesthesia, as a rite of passage; predates Islam. Pharaonic type = infibulation.

WHO classification

Type	Description
I (clitoridectomy)	Partial or total removal of the clitoris and/or prepuce
II (excision)	Clitoris and labia minora, ± labia majora
III (infibulation)	Narrowing of the introitus by cutting and joining the labia, ± clitoral excision
IV (other)	Pricking, piercing, incising, scraping, cauterising

- Older Toubia scheme numbers differ: infibulation is Type III in WHO.

Complications

- **Immediate:** pain, bleeding, shock, infection, **urinary retention**, urethral or rectal injury.
- **Urological:** recurrent **UTI** (urinary tract infection), slow stream, meatal obstruction, vaginal stone from urine stasis, **fistula** after obstructed labour.
- **Gynaecological:** retained menses, dysmenorrhoea, cysts, keloid, dyspareunia, apareunia.
- **Obstetric:** obstructed labour, tears, haemorrhage.
- **Psychological:** anxiety, depression, PTSD (post-traumatic stress disorder).

Management

- Sensitive, culturally respectful history; examine and classify; drawings or photographs for consent.
- **Deinfibulation:** simple **midline division of the scar** with subcuticular closure of each edge; exposes meatus and introitus. High success. Best before pregnancy or antenatally.
- **Never reinfibulate.**
- Excise cysts, treat stenosis, repair fistula; counselling.
- Clitoral reconstruction: less predictable, limited follow-up.

Legal aspects

- WHO and UN (United Nations) condemn FGM, including by health workers. Illegal in many countries (UK FGM Act 2003). India has no specific law.

Diagram to draw: vulva with types I–III and the deinfibulation line.

Key points to remember:

- WHO types I–IV; infibulation is type III.
- Retention, UTI, stones, fistula.
- Midline deinfibulation; never reinfibulate.

Rapid Revision Sheet – 2023-Winter Paper 1

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Principles of tissue engineering and its use in urology (LAQ)

- Triad: cells, scaffold, signals; plus bioreactor and blood supply.
- Diffusion limit about 100–200 µm: vascularity is the main barrier.
- Atala 2006 promising; phase II trial showed no gain; conduit half stenosed.
- Raya-Rivera 2011: seeded urethras in 5 boys.

Q2. Enlist various grafts used in urology and role of BMG in urology (LAQ)

- Take: imbibition 48 h, then inosculation 48 h; graft lives on its bed.
- BMG: non-keratinised, vascular lamina propria, wet-adapted.
- Avoid Stensen's duct; buccal equals lingual in success.
- Staged BMG, not skin, in lichen sclerosus.

Q3a. Management of perforated mid urethral sling (SAQ)

- Bladder trocar injury: repass laterally, continue.
- Urethral injury: abandon mesh; stage the repair.
- Late perforation: excise mesh, repair, consider autologous sling.

Q3b. Robotic radical prostatectomy (SAQ)

- Surgeon, not platform, decides outcome (Yaxley 2016).
- Interfascial plane is the standard nerve-sparing plane.
- Bladder neck stricture <2%; transfusion 1–2%.

Q3c. Kidney retrieval techniques for organ donation (SAQ)

- Cannulate, clamp, vent, cool; kidneys removed last, en bloc.
- Left kidney preferred for living donation.
- Machine perfusion reduces DGF.

Q3d. Diagnosis and management of hemorrhagic cystitis (SAQ)

- Acrolein (mesna); radiation endarteritis; BK virus.
- Exclude bladder cancer.
- Alum first; formalin after cystogram; HBOT for radiation.

Q3e. Surgical management of primary scrotal cancer (SAQ)

- Pott 1775: chimney sweeps.
- 2 cm margin; testes spared.
- Nodes decide prognosis.

Q3f. Female circumcision (SAQ)

- WHO types I–IV; infibulation is type III.
- Retention, UTI, stones, fistula.
- Midline deinfibulation; never reinfibulate.